

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Test Report

Abstract and Executive Outline

Keywords: Prompt Caching; A/B Testing; Provider Routing; Cache Affinity; Latency; Throughput; Cost Attribution; qwen3-next-80b-a3b-instruct

Abstract

This report evaluates `qwen/qwen3-next-80b-a3b-instruct` on Infron and OpenRouter across cache reuse, observed cost, throughput, E2E latency, and Streaming TTFT under Prompt Caching workloads.

The main findings are: Infron leads Cache hit rate in 0/4 routing modes; OpenRouter leads in 2/4; 2/4 tie; Infron leads Observed cost in all routing modes; Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads E2E latency in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie.

Overall, Infron's cross-mode strengths are observed cost, while OpenRouter's cross-mode strengths are throughput, Streaming TTFT, E2E latency, cache reuse. Platform choice should be driven by workload objective rather than a single headline metric.

Figure 0: Normalized Capability Radar

The five radar axes represent throughput, price, E2E latency, Streaming TTFT, and cache hit rate. Every metric is normalized to a 0–100 score, with farther outward meaning better.

Thick solid lines show platform–level contours; translucent lines and points show individual routing modes.

Conclusion Overview: Core Metric Winners by Routing Mode

Based on strict A/B paired samples. Blue represents Infron, orange represents OpenRouter; gold cells mark the winner for the routing objective.

Throughput OpenRouter wins 3/4 Max advantage 2.02%, higher is better	Observed cost Infron wins 4/4 Max advantage 98.09%, lower is better	E2E latency OpenRouter wins 3/4 Max advantage 130.58%, lower is better	Streaming TTFT OpenRouter wins 3/4 Max advantage 135.42%, lower is better	Cache hit rate Tie in 4/4 modes Max advantage 0.00%, higher is better
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Routing mode	Throughput objective	Cost objective	Latency objective	TTFT objective	Cache result
Throughput First throughput	OpenRouter advantage 2.02%	Infron advantage 98.09%	OpenRouter advantage 130.58%	OpenRouter advantage 135.42%	Tie tie
Price First price	Infron advantage 59.17%	Infron advantage 21.59%	Infron advantage 107.74%	Infron advantage 31.55%	Tie tie
Latency First latency	OpenRouter advantage 44.48%	Infron advantage 94.25%	OpenRouter advantage 9.60%	OpenRouter advantage 7.77%	OpenRouter advantage 0.00%
TTFT First ttft	OpenRouter advantage 68.94%	Infron advantage 89.62%	OpenRouter advantage 27.77%	OpenRouter advantage 28.89%	OpenRouter advantage 0.00%

Executive Outline

Dimension	Conclusion	Evidence
Controls	First/second <code>usage.prompt_tokens</code> deltas are limited to 50 tokens within each <code>sort/group/round</code> pair.	Methods and data quality
Cache reuse	Infron leads Cache hit rate in 0/4 routing modes; OpenRouter leads in 2/4; 2/4 tie	Overall metrics and mechanism section
Observed cost	Infron leads Observed cost in all routing modes	Overall metrics and provider drill-down
Performance	Infron leads Throughput in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads E2E latency in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie; Infron leads Streaming TTFT in 1/4 routing modes; OpenRouter leads in 3/4; 0/4 tie	Charts and statistical tests
Attribution boundary	Claims use observable response telemetry: usage, cost, TTFT, latency, provider fields, and cache tokens.	Provider/Route drill-down
Business meaning	Long-context, RAG-prefix, agent-tool, and high-frequency template workloads should evaluate cache rate, cost, first-token latency, and E2E latency together.	Discussion and conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Tie	Infron	Infron	Infron	Infron	Infron leads overall (4/5 metrics)
Latency First	Minimize full–response waiting time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first–token time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)

1. Introduction: Background, Questions, and Contributions

LLM inference–platform behavior is shaped by provider routing, prompt caching, streaming response handling, cost attribution, and fallback policy. This report treats the platform as an observable system and measures speed, cost, cache reuse, and first–token experience through strict A/B pairs.

1.1 Research Hypotheses

Hypothesis	Statement	Validation metric
H1	Stronger provider/cache affinity improves token–level cache hit rate for repeated stable prefixes.	Second–request cache–read tokens and token cache hit rate
H2	Higher cache hit rate can reduce observed cost, but does not necessarily reduce TTFT or E2E latency.	Observed cost, average TTFT, average request latency
H3	Different routing sorts change provider selection and produce different cost/throughput/latency frontiers.	Provider distribution, throughput, latency, cost

1.2 Contributions

- Uses response–returned `usage.prompt_tokens` as the input–token control while allowing small cross–platform accounting variance up to 50 tokens.
- Extends prompt–caching evaluation to cost, throughput, E2E latency, TTFT, provider distribution, reasoning telemetry, and paired statistical tests.
- Scopes every claim to observable response telemetry instead of private routing internals.

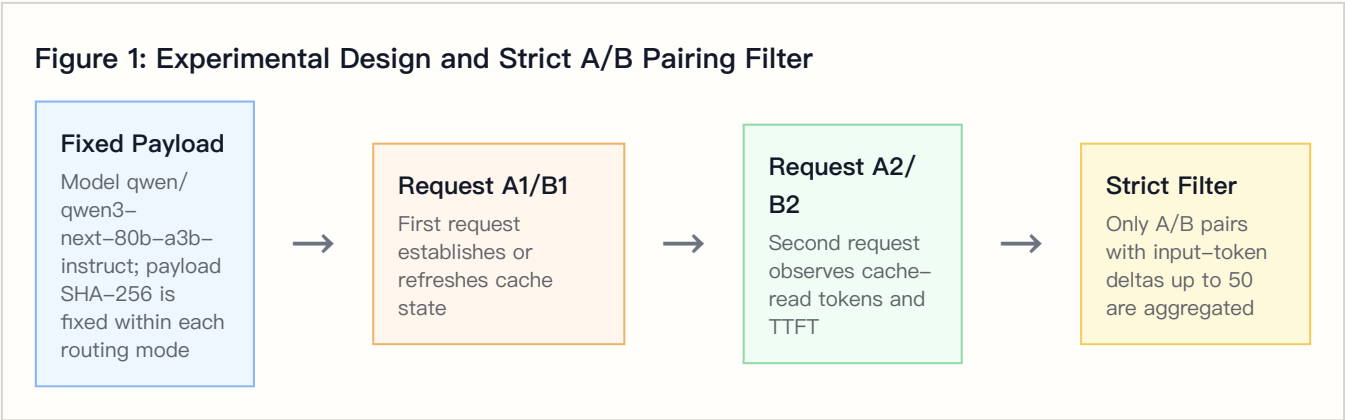
2. Experimental Design, Dataset, and Controls

2.1 Dataset Construction

The dataset is `controlled_cache_probe`, covering 4 routing sorts, 2 platforms, 4 groups, and 50 rounds per group. Each round sends identical first/second Chat Completions requests; the second request observes cache-read tokens, TTFT, and E2E latency.

The built-in business templates represent stable long-context workloads such as RAG support, agent tool instructions, marketing automation, and code review.

2.2 Controlled Variables



Controlled-variable rule: within the same `sort/group/round`, both platforms must have first/second `usage.prompt_tokens` deltas no greater than 50 tokens. Total Input Tokens are computed from response-returned usage, not local tokenizer estimates.

2.3 Metric Definitions

Metric	Definition	Direction
Call cache hit rate	Share of second requests with <code>cache_read_tokens > 0</code>	Higher is better
Token cache hit rate	Second-request cache-read tokens / second-request prompt tokens	Higher is better
Observed cost	Sum of first + second request usage/cost values	Lower is better
Throughput	Completion tokens / request latency seconds	Higher is better
E2E latency	Full response latency per request	Lower is better
TTFT	Streaming first-token arrival time	Lower is better
Reasoning telemetry	Reasoning tokens from response usage	Used to explain latency, throughput, and cost

3. Experimental Environment and Data Quality

Item	Configuration
Model	qwen/qwen3-next-80b-a3b-instruct
Provider model IDs	infron: qwen/qwen3-next-80b-a3b-instruct ; openrouter: qwen/qwen3-next-80b-a3b-instruct
Platforms	Infron and OpenRouter
API protocol	/v1/chat/completions
Routing modes	Throughput First, Price First, Latency First, TTFT First
Groups	4
Rounds per group	50
Workers	24
Request mode	Streaming Chat Completions with TTFT collection
Reasoning / thinking control	No explicit reasoning/thinking parameter; model and platform defaults are preserved
Prompt length tiers	short ≈1500, medium ≈8000, long ≈32000
Excluded records	14

4. Results: Overall Metrics and Main Findings

Routing mode	Platform	Strict pairs	Total Input Tokens	Token cache hit rate	Observed cost	Throughput	E2E latency	Streaming TTFT
Throughput First	Infron	196	6808718	0.00%	\$0.50653900	0.58 tok/s	5184.23 ms	4972.69 ms
Throughput First	OpenRouter	196	6808718	0.00%	\$1.00339906	1.75 tok/s	2248.38 ms	2112.23 ms
Price First	Infron	199	6993788	0.00%	\$0.52312900	0.95 tok/s	3155.45 ms	2988.21 ms
Price First	OpenRouter	199	6993788	0.00%	\$0.63608411	0.60 tok/s	6555.00 ms	3931.02 ms
Latency First	Infron	199	6993788	0.00%	\$0.52877600	0.95 tok/s	3190.53 ms	2949.53 ms
Latency First	OpenRouter	199	6993788	4.15%	\$1.02712370	1.37 tok/s	2911.03 ms	2736.88 ms
TTFT First	Infron	199	6993788	0.00%	\$0.52771800	0.94 tok/s	3217.63 ms	3038.02 ms
TTFT First	OpenRouter	199	6993788	8.35%	\$1.00066019	1.59 tok/s	2518.37 ms	2357.02 ms

4.1 Tail Latency and Statistical Tests

Tail percentiles expose risk that averages hide.

Routing mode	Platform	P50 latency	P95 latency	P99 latency	P50 TTFT	P95 TTFT	P99 TTFT
Throughput First	Infron	2956.76 ms	20035.53 ms	50249.60 ms	2744.43 ms	19643.82 ms	50096.43 ms
Throughput First	OpenRouter	2102.93 ms	3296.77 ms	4321.05 ms	1980.08 ms	3036.83 ms	3875.76 ms
Price First	Infron	2663.26 ms	5692.40 ms	7365.39 ms	2470.18 ms	5453.06 ms	7122.57 ms
Price First	OpenRouter	6553.93 ms	9569.13 ms	16845.84 ms	3557.95 ms	7327.31 ms	9054.06 ms
Latency First	Infron	2867.81 ms	5614.51 ms	6958.13 ms	2596.48 ms	5037.12 ms	6535.45 ms
Latency First	OpenRouter	2592.89 ms	5074.47 ms	8870.71 ms	2442.70 ms	4477.98 ms	8866.02 ms
TTFT First	Infron	2874.59 ms	5337.00 ms	7121.96 ms	2656.78 ms	4956.22 ms	6899.57 ms
TTFT First	OpenRouter	2387.41 ms	4141.73 ms	4911.39 ms	2231.37 ms	3585.24 ms	4849.19 ms

Mean deltas use bootstrap 95% CIs; p-values use paired sign-flip permutation tests.

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Throughput First	Latency: OpenRouter – Infron	–5871.72 ms	–7879.69 ms to –4134.11 ms	<0.001	196	Positive means lower Infron latency
Throughput First	TTFT: OpenRouter – Infron	–5720.93 ms	–7729.48 ms to –3985.93 ms	<0.001	196	Positive means lower Infron TTFT
Throughput First	Throughput: Infron – OpenRouter	–0.8962 tok/s	–0.9707 tok/s to –0.8253 tok/s	<0.001	196	Positive means higher Infron throughput
Throughput First	Cost: OpenRouter – Infron	\$0.00253500	\$0.00220121 to \$0.00288778	<0.001	196	Positive means lower Infron cost
Throughput First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	196	Positive means higher Infron cache hit
Price First	Latency: OpenRouter – Infron	6799.09 ms	6136.33 ms to 7459.24 ms	<0.001	199	Positive means lower Infron latency
Price First	TTFT: OpenRouter – Infron	1885.62 ms	1354.79 ms to 2385.67 ms	<0.001	199	Positive means lower Infron TTFT
Price First	Throughput: Infron – OpenRouter	0.4544 tok/s	0.4004 tok/s to 0.5084 tok/s	<0.001	199	Positive means higher Infron throughput
Price First	Cost: OpenRouter – Infron	\$0.00056761	\$0.00049651 to \$0.00064362	<0.001	199	Positive means lower Infron cost
Price First	Token Cache Hit: Infron – OpenRouter	0.00 pp	0.00 pp to 0.00 pp	1.0000	199	Positive means higher Infron cache hit
Latency First	Latency: OpenRouter – Infron	–558.98 ms	–958.23 ms to –144.12 ms	0.0047	199	Positive means lower Infron latency
Latency First	TTFT: OpenRouter – Infron	–425.31 ms	–799.83 ms to –17.22 ms	0.0275	199	Positive means lower Infron TTFT

Routing mode	Metric	Mean delta	95% CI	p-value	Pairs	Interpretation
Latency First	Throughput: Infron – OpenRouter	–0.5006 tok/s	–0.5607 tok/s to –0.4439 tok/s	<0.001	199	Positive means higher Infron throughput
Latency First	Cost: OpenRouter – Infron	\$0.00250426	\$0.00216994 to \$0.00286112	<0.001	199	Positive means lower Infron cost
Latency First	Token Cache Hit: Infron – OpenRouter	–3.27 pp	–5.90 pp to –0.98 pp	0.0162	199	Positive means higher Infron cache hit
TTFT First	Latency: OpenRouter – Infron	–1398.51 ms	–1642.75 ms to –1147.43 ms	<0.001	199	Positive means lower Infron latency
TTFT First	TTFT: OpenRouter – Infron	–1362.00 ms	–1607.44 ms to –1116.06 ms	<0.001	199	Positive means lower Infron TTFT
TTFT First	Throughput: Infron – OpenRouter	–0.6766 tok/s	–0.7220 tok/s to –0.6302 tok/s	<0.001	199	Positive means higher Infron throughput
TTFT First	Cost: OpenRouter – Infron	\$0.00237659	\$0.00203795 to \$0.00273003	<0.001	199	Positive means lower Infron cost
TTFT First	Token Cache Hit: Infron – OpenRouter	–6.96 pp	–10.60 pp to –3.69 pp	<0.001	199	Positive means higher Infron cache hit

4.2 Reasoning / Thinking Control Check

This run does not explicitly set reasoning/thinking parameters and keeps model/platform defaults; this table records reasoning telemetry under default behavior.

Routing mode	Platform	Reasoning tokens	Avg reasoning tokens/request	Reasoning requests	Avg first reasoning token	Avg TTFT	Avg E2E latency
Throughput First	Infron	0	0.0000	0	0.00 ms	4972.69 ms	5184.23 ms
Throughput First	OpenRouter	0	0.0000	0	0.00 ms	2112.23 ms	2248.38 ms
Price First	Infron	0	0.0000	0	0.00 ms	2988.21 ms	3155.45 ms
Price First	OpenRouter	0	0.0000	0	0.00 ms	3931.02 ms	6555.00 ms
Latency First	Infron	0	0.0000	0	0.00 ms	2949.53 ms	3190.53 ms
Latency First	OpenRouter	0	0.0000	0	0.00 ms	2736.88 ms	2911.03 ms
TTFT First	Infron	0	0.0000	0	0.00 ms	3038.02 ms	3217.63 ms
TTFT First	OpenRouter	0	0.0000	0	0.00 ms	2357.02 ms	2518.37 ms

4.3 API Protocol Compatibility Matrix

This run uses `/v1/chat/completions` ; this table records success response, usage, cost, and cache telemetry coverage for both platforms under that protocol.

API protocol	Endpoint	Platform	Pairs	Requests	Success rate	Usage coverage	Token usage coverage	Cost coverage	Cache telemetry coverage
chat_completions	/v1/chat/completions	Infron	800	1600	99.62%	100.00%	100.00%	100.00%	100.00%
chat_completions	/v1/chat/completions	OpenRouter	800	1600	99.94%	100.00%	100.00%	100.00%	100.00%

5. Result Visualizations by Routing Mode

The following charts are driven by the strict-paired summary data and present routing-mode, platform, cost, cache, E2E latency, Streaming TTFT, and upstream provider differences.

5.1 Core Metric Chart Overview

The charts below are generated from the same strict-paired summary data.

**Figure 3–E2:
Normalized
capability
contour**

Five metrics are normalized to 0–100, where higher is better.

**Figure 3–E3:
Cost–cache
efficiency plane**

X-axis is token cache hit rate; Y-axis is observed total cost.

**Figure
3–
E1:
Routing–
mode
core
metric
comparison**

Infron and OpenRouter are shown side by side by routing mode.

Figure 3–E4: E2E latency and Streaming TTFT

Solid lines show E2E latency; dashed lines show Streaming TTFT.

Figure 3–E5: Upstream provider distribution

Provider shares explain cache–domain and performance differences.

Routing–Mode Drill–Down Charts

The horizontal axis shows relative advantage: blue to the right indicates Infron advantage, orange to the left indicates OpenRouter advantage.

Figure 4–E1: Throughput First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E2: Price First core metric comparison

Direction and magnitude of relative advantage across five metrics.

Figure 4–E3: Latency First core metric comparison

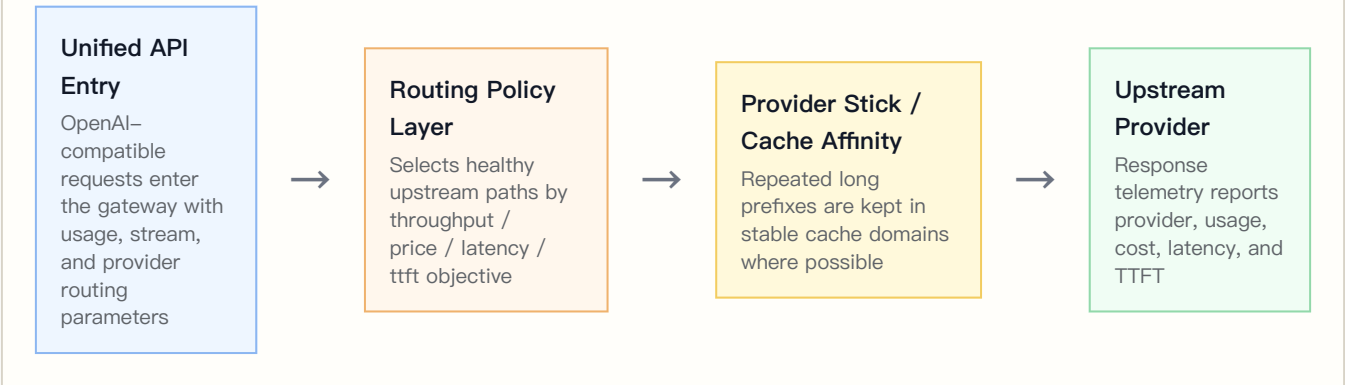
Direction and magnitude of relative advantage across five metrics.

Figure 4–E4: TTFT First core metric comparison

Direction and magnitude of relative advantage across five metrics.

6. Infron Technical Architecture and Cache/Cost Mechanism

Figure 12: Infron Multi-Provider Routing and Cache Control Plane



6.1 Multi-Provider Routing and Observable Control Plane

Requests enter a unified API, and the routing layer selects healthy upstream paths according to throughput, price, latency, or ttft objectives. Whether stable long prefixes stay in the same cache domain directly affects second-request cache-read tokens.

6.2 Provider Stick and Cache Affinity

Provider stick is cache affinity, not a permanent provider lock. Its goal is to reduce cache-domain fragmentation while respecting health and SLA constraints.

6.3 Cost-Control Path

Mechanism	Cache-rate effect	Cost effect	Observable signal
Stable prefix detection	Repeated prefixes are more likely to hit cache	Reduces repeated prefill cost	Payload SHA-256 and second-request cache-read tokens
Provider stick / cache affinity	Reduces cross-domain cache fragmentation	Avoids repeated cache warm-up	Provider distribution and token cache hit rate
Health checks and fallback	Protects availability while sometimes sacrificing cache	Reduces failure cost	HTTP status, provider distribution, tail latency
Cost-aware routing	Prefers lower-cost paths under constraints	Reduces total and per-round cost	Observed cost, cost breakdown coverage, cache-read tokens

7. Provider/Route Drill-Down

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	Infron	392	392	alibaba/sg 261, alibaba/us 75, alibaba/cn 53, deepinfra 3

Routing mode	Platform	Total requests	Attributed requests	Provider distribution
Throughput First	OpenRouter	392	392	Google 370, Alibaba 22
Price First	Infron	398	398	alibaba/sg 327, alibaba/us 66, deepinfra 5
Price First	OpenRouter	398	398	DeepInfra 371, Alibaba 27
Latency First	Infron	398	398	alibaba/sg 385, deepinfra 13
Latency First	OpenRouter	398	398	Google 384, Parasail 14
TTFT First	Infron	398	398	alibaba/sg 388, deepinfra 10
TTFT First	OpenRouter	398	398	Google 365, Parasail 33

Upstream Provider Detail Distribution

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Throughput First	Infron	alibaba/sg	261	66.58%	126/135	138	2530.39 ms	2697.00 ms	4686618	783
Throughput First	Infron	alibaba/us	75	19.13%	41/34	41	3758.29 ms	3881.56 ms	1148242	225
Throughput First	Infron	alibaba/cn	53	13.52%	29/24	36	18772.90 ms	19283.08 ms	889659	159
Throughput First	Infron	deepinfra	3	0.77%	0/3	3	4009.01 ms	5060.31 ms	84199	12
Throughput First	OpenRouter	Google	370	94.39%	185/185	194	2099.79 ms	2238.41 ms	6432792	1480
Throughput First	OpenRouter	Alibaba	22	5.61%	11/11	20	2321.36 ms	2416.05 ms	375926	66
Price First	Infron	alibaba/sg	327	82.16%	163/164	176	2748.39 ms	2914.26 ms	5791943	981
Price First	Infron	alibaba/us	66	16.58%	34/32	43	4165.89 ms	4320.57 ms	1127777	198
Price First	Infron	deepinfra	5	1.26%	2/3	5	3127.10 ms	3550.16 ms	74068	20
Price First	OpenRouter	DeepInfra	371	93.22%	183/188	199	3980.12 ms	6787.40 ms	6334107	1484
Price First	OpenRouter	Alibaba	27	6.78%	16/11	27	3256.37 ms	3361.58 ms	659681	81
Latency First	Infron	alibaba/sg	385	96.73%	195/190	198	2935.31 ms	3166.78 ms	6778119	1155

Routing mode	Platform	Upstream provider	Requests	Share	first/second	Covered rounds	Avg TTFT	Avg latency	Prompt tokens	Completion tokens
Latency First	Infron	deepinfra	13	3.27%	4/9	12	3370.71 ms	3893.94 ms	215669	52
Latency First	OpenRouter	Google	384	96.48%	193/191	196	2723.43 ms	2899.30 ms	6658786	1536
Latency First	OpenRouter	Parasail	14	3.52%	6/8	11	3105.56 ms	3232.86 ms	335002	56
TTFT First	Infron	alibaba/sg	388	97.49%	194/194	199	3007.16 ms	3170.11 ms	6848208	1164
TTFT First	Infron	deepinfra	10	2.51%	5/5	10	4235.33 ms	5061.04 ms	145580	40
TTFT First	OpenRouter	Google	365	91.71%	183/182	186	2347.05 ms	2507.22 ms	6336279	1460
TTFT First	OpenRouter	Parasail	33	8.29%	16/17	20	2467.34 ms	2641.71 ms	657509	132

7.1 Cache–Rate and Cost Divergence Drill–Down

This table combines cache, cost, provider distribution, and reasoning telemetry to explain routing–mode differences.

Routing mode	Cache–hit delta	Infron cost multiple	Infron top path	OpenRouter top path	Reasoning token delta	Main attribution
Throughput First	+0.00 pp	0.50x	alibaba/sg 66.58%	Google 94.39%	+0	Cache is tied; Infron leads cost
Price First	+0.00 pp	0.82x	alibaba/sg 82.16%	DeepInfra 93.22%	+0	Cache is tied; Infron leads cost
Latency First	–4.15 pp	0.51x	alibaba/sg 96.73%	Google 96.48%	+0	Cache and cost move in different directions; evaluate with speed metrics
TTFT First	–8.35 pp	0.53x	alibaba/sg 97.49%	Google 91.71%	+0	Cache and cost move in different directions; evaluate with speed metrics

8. Stratified Results: Prompt–Length Cache Performance

This section aggregates second–request cache–read tokens, token–level cache hit rate, observed cost, E2E latency, and Streaming TTFT by prompt–length tier. Bold cells mark the advantaged side within each tier.

Prompt–Length Tier Overview

Prompt length tier	Target tokens	Platform	Pairs	Second prompt tokens	Second cache read tokens	Token cache hit rate	Observed cost	Avg E2E latency	Avg TTFT
short	1500	Infron	268	533272	0	0.00%	\$0.08124100	2405.17 ms	2215.13 ms
short	1500	OpenRouter	268	533272	11424	2.14%	\$0.14393625	2869.80 ms	2042.66 ms
medium	8000	Infron	267	2756727	0	0.00%	\$0.41399600	3183.35 ms	2998.93 ms
medium	8000	OpenRouter	267	2756727	58752	2.13%	\$0.73276781	3541.90 ms	2662.11 ms
long	32000	Infron	258	10605042	0	0.00%	\$1.59092500	5522.19 ms	5296.33 ms
long	32000	OpenRouter	258	10605042	367200	3.46%	\$2.79056300	4305.37 ms	3688.91 ms

Prompt Length x Routing Mode Cache Hit Rate

Prompt length tier	Routing mode	Infron	OpenRouter	Winner
short	Throughput First	0.00%	0.00%	tie
short	Price First	0.00%	0.00%	tie
short	Latency First	0.00%	2.45%	OpenRouter
short	TTFT First	0.00%	6.12%	OpenRouter
medium	Throughput First	0.00%	0.00%	tie
medium	Price First	0.00%	0.00%	tie
medium	Latency First	0.00%	2.83%	OpenRouter
medium	TTFT First	0.00%	5.66%	OpenRouter
long	Throughput First	0.00%	0.00%	tie
long	Price First	0.00%	0.00%	tie
long	Latency First	0.00%	4.58%	OpenRouter
long	TTFT First	0.00%	9.16%	OpenRouter

9. Stratified Results: Group–Level Stability

Throughput First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	49	49	0.00%	\$0.11936700	36281.15 ms	36203.82 ms
Infron	2	48	48	0.00%	\$0.12747500	44657.15 ms	44514.54 ms
Infron	3	49	49	0.00%	\$0.12877800	4338.02 ms	4218.15 ms
Infron	4	50	50	0.00%	\$0.13091900	3809.54 ms	3656.68 ms
OpenRouter	1	49	49	0.00%	\$0.24587236	3498.90 ms	3387.04 ms
OpenRouter	2	48	48	0.00%	\$0.25416911	3059.15 ms	2860.55 ms
OpenRouter	3	49	49	0.00%	\$0.25315975	3189.21 ms	2904.47 ms
OpenRouter	4	50	50	0.00%	\$0.25019785	3108.18 ms	2829.80 ms

Price First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	49	49	0.00%	\$0.12383900	4669.25 ms	4152.30 ms
Infron	2	50	50	0.00%	\$0.13310900	6510.02 ms	6154.40 ms
Infron	3	50	50	0.00%	\$0.13588100	4930.64 ms	4180.06 ms
Infron	4	50	50	0.00%	\$0.13030000	4333.30 ms	4217.16 ms
OpenRouter	1	49	49	0.00%	\$0.14909698	16772.30 ms	6710.55 ms
OpenRouter	2	50	50	0.00%	\$0.16423321	9564.35 ms	7229.66 ms
OpenRouter	3	50	50	0.00%	\$0.16407179	9037.89 ms	7185.26 ms
OpenRouter	4	50	50	0.00%	\$0.15868213	9067.03 ms	8462.85 ms

Latency First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	49	49	0.00%	\$0.12590300	4579.91 ms	4346.15 ms
Infron	2	50	50	0.00%	\$0.13765700	5188.43 ms	4807.22 ms
Infron	3	50	50	0.00%	\$0.13491600	5539.01 ms	5369.55 ms
Infron	4	50	50	0.00%	\$0.13030000	5939.77 ms	5619.08 ms
OpenRouter	1	49	49	0.00%	\$0.24814680	6887.25 ms	6453.01 ms
OpenRouter	2	50	50	4.50%	\$0.26698700	4688.96 ms	4440.95 ms
OpenRouter	3	50	50	6.00%	\$0.25958222	3830.05 ms	3612.86 ms

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
OpenRouter	4	50	50	5.84%	\$0.25240768	4725.16 ms	4239.98 ms

TTFT First

Platform	Group	Rounds	Successful rounds	Token cache hit rate	Observed cost	P95 latency	P95 TTFT
Infron	1	50	50	0.00%	\$0.13030000	5178.89 ms	4599.93 ms
Infron	2	49	49	0.00%	\$0.13129800	5831.73 ms	5283.54 ms
Infron	3	50	50	0.00%	\$0.13520100	4810.97 ms	4573.64 ms
Infron	4	50	50	0.00%	\$0.13091900	5478.90 ms	5353.08 ms
OpenRouter	1	50	50	5.84%	\$0.25137438	4190.14 ms	3487.44 ms
OpenRouter	2	49	49	6.04%	\$0.25145242	3770.40 ms	3562.88 ms
OpenRouter	3	50	50	10.91%	\$0.25198491	3929.62 ms	3611.54 ms
OpenRouter	4	50	50	10.54%	\$0.24584848	4061.76 ms	3491.50 ms

10. Discussion: Business Value, Boundaries, and Engineering Implications

Business decisions should not rely on one metric. Stable long-context and high-frequency template workloads should prioritize cache rate and cost; realtime interaction must constrain TTFT and E2E latency; batch processing often prioritizes throughput and failure cost.

Routing mode	Business objective	Observed result	Scenarios	Caveat
Throughput First	Maximize output capacity per unit time	OpenRouter leads overall (3/5 metrics)	Batch generation, offline summaries, backend processing	First-token and E2E response are faster; check cache and cost
Price First	Minimize request and token cost	Infron leads overall (4/5 metrics)	High-frequency templates, support automation, marketing, RAG prefixes	Decide with budget, SLA, and cache stability together
Latency First	Minimize full-response waiting time	OpenRouter leads overall (4/5 metrics)	Online chat, agent chains, IDE/writing assistants, realtime tools	First-token and E2E response are faster; check cache and cost
TTFT First	Minimize streaming first-token time	OpenRouter leads overall (4/5 metrics)	Streaming chat, realtime copilots, first-screen feedback	First-token and E2E response are faster; check cache and cost

11. Conclusion

Routing–Mode Conclusions

Routing mode	Objective	Cache winner	Cost winner	Throughput winner	E2E latency winner	TTFT winner	Interpretation
Throughput First	Maximize output capacity per unit time	Tie	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (3/5 metrics)
Price First	Minimize request and token cost	Tie	Infron	Infron	Infron	Infron	Infron leads overall (4/5 metrics)
Latency First	Minimize full–response waiting time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)
TTFT First	Minimize streaming first–token time	OpenRouter	Infron	OpenRouter	OpenRouter	OpenRouter	OpenRouter leads overall (4/5 metrics)

12. Limitations, Missing Data, and Next Experiments

Limitation	Impact	Next step	Current handling
Full routing trace	Cannot prove every provider choice, fallback, and retry path hop by hop	Add provider routing trace, decision logs, and fallback reasons	Use only returned provider fields and provider distribution
Longer time window	4x50 observes short–window stability but not day–level drift	Add soak tests and repeated windows	Scope conclusions to this run
Production corpus	Built–in templates do not cover every workload distribution	Use sanitized production–stratified corpora	Discuss representative long–context templates only
Cost–field consistency	Cost coverage and semantics may differ by platform	Reconcile with billing and provider cost breakdown	Use only explicitly returned cost fields

13. Reproducibility Appendix

Artifact	Path
Summary	export/qwen3_next_80b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions summary.json
Paired dataset	export/qwen3_next_80b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_pairs.csv

Artifact	Path
Request-level dataset	export/qwen3_next_80b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions benchmark_requests.jsonl
Filtered structured records	export/qwen3_next_80b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records.json
Excluded-record audit	export/qwen3_next_80b_all_experiments/ routing_sort_cache_cost_ab_4x50_stream_ttft_reasoning_default_length_tiers_chat_completions records_excluded.json
Test source	tests/test_rerun_routing_sort_cache_cost_ab.py
Benchmark runner source	scripts/rerun_routing_sort_cache_cost_ab.py
HTML report renderer source	scripts/render_glm52_deepseek_style_report.py
Dataset reference	business_representative built-in representative business templates; request-level export is benchmark_requests